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Java SE 27 & JDK 27

Class Base64

java.lang.Object
    java.util.Base64

```
public final class Base64
extends Object
```

This class consists exclusively of static methods for obtaining encoders and decoders for the Base64 encoding scheme. The implementation of this class supports the following types of Base64 as specified in RFC 4648 and RFC 2045.

- **Basic**

  Uses "The Base64 Alphabet" as specified in Table 1 of RFC 4648 and RFC 2045 for encoding and decoding operation. The encoder does not add any line feed (line separator) character. The decoder rejects data that contains characters outside the base64 alphabet.

- **URL and Filename safe**

  Uses the "URL and Filename safe Base64 Alphabet" as specified in Table 2 of RFC 4648 for encoding and decoding. The encoder does not add any line feed (line separator) character. The decoder rejects data that contains characters outside the base64 alphabet.

- **MIME**

  Uses "The Base64 Alphabet" as specified in Table 1 of RFC 2045 for encoding and decoding operation. The encoded output must be represented in lines of no more than 76 characters each and uses a carriage return `'\r'` followed immediately by a linefeed `'\n'` as the line separator. No line separator is added to the end of the encoded output. All line separators or other characters not found in the base64 alphabet table are ignored in decoding operation.

Unless otherwise noted, passing a `null` argument to a method of this class will cause a `NullPointerException` to be thrown.

Since:

1.8

External Specifications

RFC 2045: Multipurpose Internet Mail Extensions (MIME) Part One: Format of Internet Message Bodies,

RFC 4648: The Base16, Base32, and Base64 Data Encodings

Nested Class Summary

Nested Classes

Modifier and Type	Class	Description
static class	Base64.Decoder	This class implements a decoder for decoding byte data using the Base64 encoding scheme as specified in RFC 4648 and RFC 2045.
static class	Base64.Encoder	This class implements an encoder for encoding byte data using the Base64 encoding scheme as specified in RFC 4648 and RFC 2045.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
static Base64.Decoder	getDecoder()	Returns a Base64.Decoder that decodes using the Basic type base64 encoding scheme.
static Base64.Encoder	getEncoder()	Returns a Base64.Encoder that encodes using the Basic type base64 encoding scheme.
static Base64.Decoder	getMimeDecoder()	Returns a Base64.Decoder that decodes using the MIME type base64 decoding scheme.
	getMimeEncoder()	

static Base64.Encoder		Returns a Base64.Encoder that encodes using the MIME type base64 encoding scheme.
static Base64.Encoder	**getMimeEncoder**(int lineLength, byte[] lineSeparator)	Returns a Base64.Encoder that encodes using the MIME type base64 encoding scheme with specified line length and line separators.
static Base64.Decoder	**getUrlDecoder**()	Returns a Base64.Decoder that decodes using the URL and Filename safe type base64 encoding scheme.
static Base64.Encoder	**getUrlEncoder**()	Returns a Base64.Encoder that encodes using the URL and Filename safe type base64 encoding scheme.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Details

getEncoder

```
public static Base64.Encoder getEncoder()
```

Returns a Base64.Encoder that encodes using the Basic type base64 encoding scheme.

Returns:

A Base64 encoder.

getUrlEncoder

```
public static Base64.Encoder getUrlEncoder()
```

Returns a `Base64.Encoder` that encodes using the URL and Filename safe type base64 encoding scheme.

Returns:

A Base64 encoder.

getMimeEncoder

```
public static Base64.Encoder getMimeEncoder()
```

Returns a `Base64.Encoder` that encodes using the MIME type base64 encoding scheme.

Returns:

A Base64 encoder.

getMimeEncoder

```
public static Base64.Encoder getMimeEncoder(int lineLength,
                                            byte[] lineSeparator)
```

Returns a `Base64.Encoder` that encodes using the MIME type base64 encoding scheme with specified line length and line separators.

Parameters:

`lineLength` - the length of each output line (rounded down to nearest multiple of 4). If the rounded down line length is not a positive value, the output will not be separated in lines

`lineSeparator` - the line separator for each output line

Returns:

A Base64 encoder.

Throws:

`IllegalArgumentException` - if `lineSeparator` includes any character of "The Base64 Alphabet" as specified in Table 1 of RFC 2045.

getDecoder

```
public static Base64.Decoder getDecoder()
```

Returns a `Base64.Decoder` that decodes using the Basic type base64 encoding scheme.

Returns:

A Base64 decoder.

getUrlDecoder

```
public static Base64.Decoder getUrlDecoder()
```

Returns a `Base64.Decoder` that decodes using the URL and Filename safe type base64 encoding scheme.

Returns:

A Base64 decoder.

getMimeDecoder

```
public static Base64.Decoder getMimeDecoder()
```

Returns a `Base64.Decoder` that decodes using the MIME type base64 decoding scheme.

Returns:

A Base64 decoder.

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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